



SWE 3513

Artificial Intelligence

## AI\_A1\_G14 UI and UX design

### Musanze Cooperative Harvest and Dispatch Decision Dashboard

*A proposed staff interface for HarvestLink operations · Group AI-G14*

| Document detail  | Content                                                                                                                                                                                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group code       | AI-G14                                                                                                                                                                                                                                                    |
| Group leader     | Teta Erssie                                                                                                                                                                                                                                               |
| Members          | Joel Eliezer Youto Mongar Jr (Member 1, Data and UX); Niyibigira Gad (Member 2, Regression); Uwiringiyimana Marie Claire (Member 3, Classification); Mbabazi Sandrine (Member 4, Clustering and QA); Teta Erssie (Member 5, Reproducibility and release). |
| Stakeholder      | HarvestLink operations team: collection clerks, dispatchers, and the floor supervisor who authorises trucks.                                                                                                                                              |
| Document type    | Five-to-seven page staff dashboard proposal. The interface is not implemented in code.                                                                                                                                                                    |
| Related commands | <pre>python run_all.py --data data/AI_A1_G14.csv --output artifacts/ --group AI-G14 python predict.py --record '{...}'</pre>                                                                                                                              |

### Cover and operational problem

Musanze HarvestLink Cooperative collects potatoes from several fictional farms. Before a truck leaves, non-technical staff must act on three model outputs from the same record. This PDF shows how those outputs appear on a proposed dashboard. It does not replace the command-line pipeline.

| Decision                                                | Model output the staff see                                                                          | Action the staff take                                                                        |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>1. Estimate expected harvest weight</b>              | Predicted actual_yield_kg from NumPy linear regression, shown in kilograms with last test MAE/RMSE. | Plan crates and truck space. If confidence is low, weigh the load before dispatch.           |
| <b>2. Flag consignments needing dispatch attention</b>  | Predicted dispatch_attention (0 or 1) and the probability of class 1 from logistic regression.      | Hold class-1 loads for a supervisor check. A missed flag is more costly than an extra check. |
| <b>3. Group collection points with similar profiles</b> | Cluster label from k-means on input features only, with a caution that clusters are not farm types. | Compare similar collection points. Do not treat a colour or number as a verified category.   |



## User journey and information flow

The journey starts with the lecturer-issued CSV and ends with a human decision. No truck is released by the model. The dashboard is a reading layer over `run_all.py` and `predict.py`.

| Step                     | User action                                                   | System response                                                                                          | Human decision                                                                           |
|--------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>1. CSV upload</b>     | Clerk selects data/AI_A1_G14.csv (schema unchanged).          | Pipeline checks columns and types, reports missing values and duplicates, prints group code and SHA-256. | Accept the file only if the fingerprint and schema match. Reject renamed or mixed files. |
| <b>2. Quality review</b> | Clerk opens the data-quality screen.                          | Shows row count, feature count, missingness, and that <code>record_id</code> is not a model feature.     | Stop if the schema fails. Continue only when the quality gate is green.                  |
| <b>3. Training run</b>   | Analyst runs <code>run_all.py</code> from the README command. | Writes metrics, plots, and models into artifacts/ and models/. Overwrites previous outputs.              | Read MAE, confusion matrix, and selected k. Do not type metrics by hand.                 |
| <b>4. Prediction</b>     | Dispatcher enters one farm record (six input fields).         | <code>predict.py</code> returns JSON: yield kg, class, probability, cluster, group code, model version.  | Read the numbers on the prediction screen. Incomplete JSON is blocked, not filled in.    |
| <b>5. Human decision</b> | Supervisor chooses Confirm, Hold, or Override.                | Stores the human choice with the model advice attached.                                                  | The supervisor's decision is authoritative. Class 1 never auto-dispatches.               |

## Information flow (summary)

CSV file → schema and SHA-256 gate → feature matrix (identifiers and targets removed) → regression / classification / clustering artefacts → single-record JSON → staff review → Confirm / Hold / Override. Targets `actual_yield_kg` and `dispatch_attention` are never clustering inputs. `record_id` is never a model feature.



## Annotated wireframes

Pages 3 and 4 show the four required screens: data quality, prediction, classification warning, and clusters. Callouts in the tables under each figure state what a non-technical user should do.

**Figure 1. Data quality review**

| HarvestLink   Data quality   Group AI-G14                                                                                                          |                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Dataset fingerprint (SHA-256)</b><br>Shown in full and copied to Moodle. Must match data_report.json.                                           |                                                                                             |
| <b>Schema checklist</b><br>record_id, plot_area_ha, rainfall_mm, soil_ph, seed_kg, distance_km, arrival_hour, actual_yield_kg, dispatch_attention. |                                                                                             |
| <b>Counts (computed, not typed)</b><br>Row count · Feature count · Missing values · Duplicate rows · Duplicate record_id                           |                                                                                             |
| <b>Gate</b><br>[ Reject file — schema mismatch ] [ Continue to models ]<br>If columns differ, training is blocked.                                 |                                                                                             |
| Callout                                                                                                                                            | What the user understands                                                                   |
| <b>A. Fingerprint</b>                                                                                                                              | This is the same file the lecturer issued. A different hash means a replaced or edited CSV. |
| <b>B. Schema list</b>                                                                                                                              | Only the published columns are allowed. Extra or renamed fields stop the run.               |
| <b>C. Counts</b>                                                                                                                                   | Quality numbers come from the pipeline. Staff do not paste metrics into the screen.         |
| <b>D. Gate</b>                                                                                                                                     | Prediction is unavailable until the file passes. This prevents acting on a broken upload.   |

**Figure 2. Harvest prediction**

| HarvestLink   Score one consignment   predict.py                                                                                                                                          |                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Input fields</b> plot_area_ha rainfall_mm soil_ph seed_kg distance_km arrival_hour                                                                                                     |                                                                                |
| <b>Command</b> python predict.py --record '{...six fields...}'                                                                                                                            |                                                                                |
| <b>Returned JSON (read-only)</b><br>regression_prediction_kg<br>classification_prediction + classification_probability<br>cluster_label · group_code · model_version                      |                                                                                |
| <b>Staff reading</b><br>Predicted yield: [ kg ] Last test MAE / RMSE shown beside the number.<br>Low R <sup>2</sup> or a very small test set: display "low confidence — check the scale". |                                                                                |
| Callout                                                                                                                                                                                   | What the user understands                                                      |
| <b>E. Six inputs only</b>                                                                                                                                                                 | The dispatcher types farm measurements. record_id is not required for scoring. |
| <b>F. JSON contract</b>                                                                                                                                                                   | The screen shows the same keys the assessor will see in the terminal.          |
| <b>G. kg plus error</b>                                                                                                                                                                   | A point estimate without MAE/RMSE would look more precise than the model is.   |



Figure 3. Classification warning

| HarvestLink   Dispatch attention                                                                                                                                                              |                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>WARNING — ATTENTION REQUIRED</b><br>Predicted class: 1 Probability of attention: shown as 0–1, not as a percentage slogan.                                                                 |                                                                                                |
| <b>Cost rule shown on screen</b><br>False negative (missed class 1) is more costly than false positive (extra check).<br>Class 1 = hold for supervisor. Class 0 = proceed with normal checks. |                                                                                                |
| <b>Last training confusion matrix</b><br>TN   FP<br>FN (costly)   TP                                                                                                                          |                                                                                                |
| <b>Actions</b> [ Hold truck ] [ Confirm class 0 ] [ Override — reason required ]                                                                                                              |                                                                                                |
| Callout                                                                                                                                                                                       | What the user understands                                                                      |
| <b>H. Class and probability together</b>                                                                                                                                                      | 0.51 is not the same as 0.95. Borderline scores invite Hold, not autopilot.                    |
| <b>I. FN highlighted</b>                                                                                                                                                                      | Missing a load that needed attention can send an at-risk consignment. Extra review is cheaper. |
| <b>J. Override</b>                                                                                                                                                                            | The model advises. A named supervisor records why the advice was not followed.                 |

Figure 4. Collection-point clusters

| HarvestLink   Operating profiles   unsupervised k-means                                                    |                                                                           |
|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Plot</b> PCA view of input features only. Colour = selected k from silhouette (k compared from 2 to 5). |                                                                           |
| <b>Legend</b> Cluster 0, Cluster 1, ... Label: "similar input profile", not "good farm" or "bad farm".     |                                                                           |
| <b>On-screen caution</b> Clusters are groupings in this file. They are not verified real-world categories. |                                                                           |
| <b>New record</b> Assigned to the nearest centroid. No target columns are used.                            |                                                                           |
| Callout                                                                                                    | What the user understands                                                 |
| <b>K. Similarity only</b>                                                                                  | Nearby points share plot, rain, pH, seed, distance, and arrival patterns. |
| <b>L. No ranking</b>                                                                                       | A cluster number is not a performance grade and must not be used as one.  |
| <b>M. Targets excluded</b>                                                                                 | actual_yield_kg and dispatch_attention are not clustering inputs.         |



## Responsible AI states

The dashboard must remain usable when the model is uncertain, when data are missing, when the method is limited, and when a person disagrees with the score. These four states are required evidence for this page.

| State                            | What the staff see                                                                                                                                | What the system does                                                                                                                |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Uncertainty</b>               | Predicted kg with MAE/RMSE. Attention shown as class plus probability. Banner "low confidence" when test $R^2$ is weak or the test split is tiny. | Does not hide intervals or probabilities. Does not present a single kilogram as a certified weight.                                 |
| <b>Missing or malformed data</b> | Red list of missing field names, or "Malformed JSON". No invented defaults for soil_ph or other inputs.                                           | predict.py rejects the record and returns a JSON error. Training stops if the CSV schema does not match the published columns.      |
| <b>Model limitation</b>          | Plain-language note: linear yield, logistic attention, k-means profiles. "This is advice, not a farm verdict."                                    | record_id is never a feature. Supervised scalars fit on training rows only. Hidden assessor CSVs must recompute metrics.            |
| <b>Human override</b>            | Buttons Confirm, Hold, Override. Override requires a short reason (weather, late truck, damaged crate).                                           | Stores the human decision as the official outcome. The model output remains attached as advice. No automatic dispatch from class 1. |

Live verification note. Named constants live in `src/config.py` (RANDOM\_SEED, LEARNING\_RATE, N\_ITERATIONS, K\_MIN, K\_MAX). A member can change one value and explain the new artefacts without editing hidden row counts.



## Visual system and rationale

Colour, labels, and legends follow the same navy-and-table language as this assignment document so staff are not asked to learn a separate brand. Colour is never the only signal.

| Element                               | Specification                                                                         | Accessibility reason                                                                      |
|---------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Primary navy #17365D</b>           | Screen titles, table headers, primary buttons.                                        | High contrast on white. Matches the official INES table header in this assignment format. |
| <b>Alert amber #C43E00 on #FFF4E5</b> | Dispatch-attention banner and false-negative cell.                                    | Not red-only: the banner also contains the word REQUIRED and the class number.            |
| <b>Body text black on white</b>       | All reading text, including kg values.                                                | Contrast above typical 7:1. No light-grey body copy.                                      |
| <b>Labels</b>                         | "Predicted kg", "Attention probability", "Cluster (unsupervised)".                    | Verbs on buttons: Upload, Score, Hold, Override. No "safe farm" label.                    |
| <b>Legends</b>                        | Cluster colours named Cluster 0, Cluster 1, with the caution sentence under the plot. | Colour-blind users still have the numeric label and the caution text.                     |

## Three justified product decisions

| Decision                                                  | Justification                                                                                                                                                                                                            |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Show kilograms together with MAE/RMSE</b>           | A lone predicted kilogram invites over-trust. Staff plan crates from a number plus an error scale, then weigh when confidence is low.                                                                                    |
| <b>2. Pair class 1 with probability and a Hold action</b> | A binary flag hides borderline cases. Probability plus Hold/Override keeps a person in the loop when the score is near 0.5. False negatives remain the costly error.                                                     |
| <b>3. Name clusters as profiles, never as grades</b>      | Silhouette only ranks compactness of groups in feature space. Calling a cluster poor or rich would be a claim the method cannot support, and the assignment forbids treating clusters as verified real-world categories. |

## Page checklist against the assignment table

| Page          | Required content                  | Where it is in this PDF                                      |
|---------------|-----------------------------------|--------------------------------------------------------------|
| <b>1</b>      | Cover and operational problem     | Group code, members, stakeholder, three decisions.           |
| <b>2</b>      | User journey and information flow | CSV upload → review → prediction → human decision.           |
| <b>3 to 4</b> | Annotated wireframes              | Data quality, prediction, classification warning, clusters.  |
| <b>5</b>      | Responsible AI states             | Uncertainty, missing data, model limitation, human override. |
| <b>6</b>      | Visual system and rationale       | Colours, labels, legends, three justified decisions.         |